

BenGER Transformation: Exploring Automated Rubric Generation to Reduce LLM Grading Variance in German Legal Exams

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LLM judges can grade free-text legal exam solutions at scale, but their scores vary across judges and across repeated runs. Careful examiners reduce such variance with a grading sheet (*Bewertungsbogen*) that splits the expected solution into small steps worth a fixed number of points. We ask whether a sheet generated automatically from the case and its model solution reduces the variance of LLM grading of German law exams, and at what cost to agreement with human graders. In a first iteration, twelve models generated sheets for 15 exams, and judges re-graded 45 answers with blind human grades. The sheets did not make capable judges steadier on average: the effect depended on the judge, the sheets cost some agreement with the human graders, and under the sheet several judges gave high scores to empty or copied answers. We then rebuilt the pipeline. The model writes small steps with requirements, the judge must support each credit with a quote from the answer, code allocates the points and checks every quote, and defensible alternative solutions and auxiliary opinions are represented explicitly. We validate it against grades of real exam scripts on the exam author's own sheet, benchmark which models generate good sheets and which use them well, and describe how such exams are graded in practice. **[PENDING: results of the full benchmark and the recommended pipeline]**

CCS Concepts: • **Applied computing** → **Law; Education**; • **Computing methodologies** → *Natural language processing*.

Additional Key Words and Phrases: LLM-as-a-judge, grading rubrics, legal education, German law examinations, automated assessment, reliability and validity

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1 Introduction

German law students learn and are examined through case solutions (*Falllösungen*). A candidate receives a set of facts (*Sachverhalt*) and writes a legal opinion in a prescribed appraisal style (*Gutachtenstil*). The state examinations grade such opinions on a statutory scale from 0 to 18 points [6]. Grading them is hard. Different examiners give the same script different grades [21]. To grade consistently, correctors often build a point scheme over the expected solution before they grade [15]. Many do this with a grading sheet (*Bewertungsbogen*): the expected solution split into small steps, each worth a number of points (*Bewertungseinheiten*, BE), and a key that maps the BE total to a grade.

Large language models (LLMs) can grade such texts at scale. On German case solutions, an LLM judge with one rubric for all exams tracks human graders on average, but its scores vary between judges and between repeated runs [34]. This paper studies the obvious remedy: let one model write an exam-specific grading sheet from the case and its model solution (*Musterlösung*), and let a judge

model grade against it. If it works, the sheet becomes an auditable, editable artifact between the model solution and the grade, which an examiner can read and correct.

A first iteration tested this remedy with a rich step-score sheet on 15 exams with 45 answers and blind human grades. The sheets did not make capable judges steadier on average, they cost some agreement with the human graders, and under the sheet several judges gave high scores to empty or copied answers (Section 6.1). We rebuilt the pipeline and ask the question again, now also asking *which* models write good sheets and *which* use them well, against the strongest yardstick we can get: grades of real exam scripts on the exam author’s own sheet and scale. RQ1 and RQ2 carry the story; the others explain the result.

- **RQ1 Variance.** Does grading against a generated sheet make judges steadier, across repeated passes, across judges and across independently generated sheets for the same exam?
- **RQ2 Validity.** How close are sheet-based grades to human grades, and to the exam author’s grades on his own sheet and scale?
- **RQ3 Generation.** Do heterogeneous generators produce compliant, legally sound, exam-specific sheets under one output specification?
- **RQ4 Generator benchmark.** Which generators write the sheets that let a fixed judge grade most validly and reliably, and do cheap signals predict this?
- **RQ5 Judge benchmark.** Which judges use a fixed sheet best, in validity, stability, robustness to degenerate answers, and cost?
- **RQ6 Pipeline.** Which combination of design choices gives the lowest error against the expert within a cost limit?
- **RQ7 Grading practice.** How do examiners distribute points, what do they credit, where do they set the pass mark, how do they map points to grades, and how do they treat cardinal errors (*Kardinalfehler*)?

Our contributions are:

1. A grading-sheet pipeline in which the model specifies and code allocates: half-point BE, requirement bullets with a standard of assessment, structural forks with alternative solution paths, owed auxiliary opinions, and quotes verified by code (Section 4).
2. A generator benchmark with 12 generators on 16 exams, measuring sheet-sample variance and validity against human and expert grades (Section 5).
3. A judge benchmark with 6 to 8 judges on fixed sheets, measuring validity, repeat variance, robustness to probes, and cost.
4. The interaction of generator and judge, and a recommended pipeline confirmed once on held-out expert grades.
5. An account of how German law exams are graded, co-written with an exam author (Section 2), and the defects of the first-iteration pipeline that the rebuild fixes (Section 6.1, Appendix A).

[DRAFT: contributions 2 to 4 describe experiments E1 to E4, which have not run yet; see Section 6]

The study is exploratory and was not pre-registered. Every run records the hashes of its code, prompts and validator, and we report every run we make.

2 Grading practice in German law exams

[DRAFT: to be co-written with the exam author; every pending item below needs his input or experiment E6/E7]

The setting. The statutory scale has 7 named bands from 0 to 18 points; 4 points (*ausreichend*) is the lowest pass grade [6]. Examiners in the state examinations grade holistically. Many first write down the expected solution and what each part is worth [15]. University exams often make this explicit: a sheet assigns BE to steps, and a key maps the BE total to the scale. Two principles shape

Table 1. Two expert grading paradigms and our instrument. D2: the exam author’s sheet; Z2: Gregor Roth’s sheet and workbook (we hold no case or scripts). ? = to be completed with the examiners. [PENDING: rows on alternatives and result items, confirmed with both examiners]

	Exam author (D2)	Roth (Z2)	Our instrument
Unit of credit	absolute BE	relative share per level	BE, allocated by code
Total	100 BE	100%	100 BE
Leaves	46 steps	40 leaves, up to 4 levels deep	steps with 1–6 bullets
Leaf scoring	BE on a 0.5 grid	grade 0–18, whole points	bullet status, step BE, or grade
Largest step	10 BE	9% of the grade	cap 10 BE
Aggregation	sum of BE	weighted mean of grades	sum of BE (both for grades)
Pass mark	40% of BE	grade 4	the exam’s key
Adjustment	?	manual bonus or malus	none
Result items	none; some titles	1 of 4 weighted	none; with its reasoning
Graders per script	one (see Section 5)	? (2 final-grade fields)	judge, <i>k</i> passes
Alternatives	?	?	bullets or forks

every sheet. A tenable solution (*vertretbare Lösung*) that is argued with sound reasons must not be marked as wrong [7]. And a consequential error (*Folgefehler*) is not penalised twice: a later step that correctly builds on the candidate’s own earlier decision still earns credit.

Two expert paradigms. We compare two expert sheets with our instrument (Table 1). The exam author’s sheet (D2) is a list of steps with absolute BE, grouped into sections. Gregor Roth’s sheet for a civil-law exam (Z2) is hierarchical: its weighted items state their share of the parent item, and each of the 40 weighted leaves is graded in whole points from 0 to 18. His grading workbook multiplies the shares down to absolute weights and averages the leaf grades with them; its legend marks the one place where it adjusted the stated shares. Our instrument keeps the half-point BE of the first paradigm, adds requirement bullets, lets code do the arithmetic, and can elicit the weights in the way of either paradigm (Section 4).

Q1: How should the BE be distributed? The author’s sheet spreads 100 BE over 46 steps with at most 10 BE per step. He re-weighted it between runs: the sheet exists in 3 states, and the earliest differs from the later ones in the maxima of several steps [PENDING: exam author to confirm what changed between the states]. We took the step-size guidance of our instrument (0.5 to 4 BE per step, at most 10 BE) from the two expert sheets, so step size is an input of our design, not a finding. E6 compares the BE distributions of generated sheets with both expert sheets. [PENDING: BE distribution of the expert sheet and of generated sheets (E6)]

Q2: What earns credit, including method? In the author’s sheet every step credits an issue spotted or an argument made. No step credits a bare result; some step titles state a sub-result instead. Our generator follows the same rule: no step may credit only an intermediate or final result: a result earns credit only together with the reasoning that carries it, and where several results are tenable the bullets credit the argument, not the result. Whether method (structure, appraisal style, precise language) earns separate credit is open [PENDING: exam author]. The first iteration kept style outside the points as a separate construct. E6 compares sheets with and without method bullets.

Q3: Where is the pass mark? The author’s key passes a script at 40% of the BE. The platform’s default key, which the first-iteration holistic instrument also used, passes at 50%. The pass mark decides pass/fail agreement between a judge and the examiner. E6 sweeps it from 35% to 55% on stored outputs. [PENDING: E6]

Q4: Is the key linear or curved? The author’s key is piecewise linear with a kink at the pass mark: below it a grade point spans 10 BE, from the pass mark upwards 4 BE, up to grade 18 at 96 BE. [PENDING: exam author: why this shape] The platform’s default key has the same shape: below

the pass mark a grade point spans 12.5 percentage points of the total on average, from the pass mark upwards 3.3. Roth’s paradigm needs no key: it averages statutory grades and passes from grade 4. Our rating unit supports both views: a leaf grade maps back to BE through the exam’s own key, and the weighted mean of the leaf grades is reported next to it.

Q5: How are cardinal errors treated? Some errors are grave enough that examiners limit the grade as a whole. [PENDING: exam author: definition, examples and expected loss] Our instrument has no such cap yet. The consequential-error rule and the standard of each bullet decide how far an early error propagates (Section 4.6). E6 runs 5 probes with the losses the author expects.

3 Related work

Judge reliability. Direct-scoring LLM judges [30, 46] correlate with humans on average but are unreliable at the instance level: scores drift across runs [19], flip under reordering and prompt changes [38, 42], show self-preference [35], and need validation against humans for each task [4, 17]. Most remedies act on the judge, such as calibration or panels [41]. We act on the instrument.

Rubrics and checklists. Fine-grained decomposition can make evaluation more reliable: skill-wise scoring, rubric-conditioned evaluators, atomic facts, and binary checklists that an LLM can generate per instance [8, 11, 18, 20, 25, 28, 32, 43, 45]. These systems use binary criteria or categorical importance labels with arithmetic in code; to our knowledge none asks the model for an exact point total, which LLMs handle poorly [2, 44]. Educational measurement finds that analytic rubrics are more reliable than holistic scoring at a cost in rater time [5, 22, 26]; an LLM rater turns that cost into inference cost. LLM essay scoring has used fixed, exam-independent rubrics [24, 33, 40], and prior decompositions carry uniform or binary weights and no doctrine of partial credit [27]. Work on attribution checks whether generated text is supported by cited sources [16]; our verifier asks a narrower question, whether a quote stands in the answer at all.

Grading legal free text. Legal NLP has moved from LLMs taking exams [10, 23] to LLMs grading them with per-question structure: reasoning guidance, span-level criteria, expert-written or official rubrics, and atomic judging [1, 9, 12–14, 36, 37, 39]. All of them assume an expert-written scheme. To our knowledge, no prior work generates the exam-specific point scheme, turns tenable alternative solutions into scoreable paths, and validates the result against the exam’s author on his own sheet and scale. Empirical work on German exam grading documents low agreement between examiners [21] and describes how examiners should correct [15]; we add a model of the grading sheet that makes these practices explicit.

4 The pipeline

[DRAFT: describes the implementation of output specification checklist-3; the instrument is frozen only after the probe gate E0, and this section is updated then]

The pipeline has five stages (Figure 1). Models do two things only. The generator states the content of the sheet and relative weights, and the judge marks requirements and quotes the answer. Code does everything that can be computed or checked: validation, allocation, rendering, quote verification and aggregation. The first iteration showed why. When an earlier output specification asked the model for exact point totals, almost no generator produced a valid sheet [PENDING: yield of the exact-arithmetic specification, from the withheld sweep log].

4.1 Case preparation

Each exam provides a case text and a model solution. We clean both: markup escapes and tables of contents are removed, footnotes are kept. A script is graded against the case text of its own year.

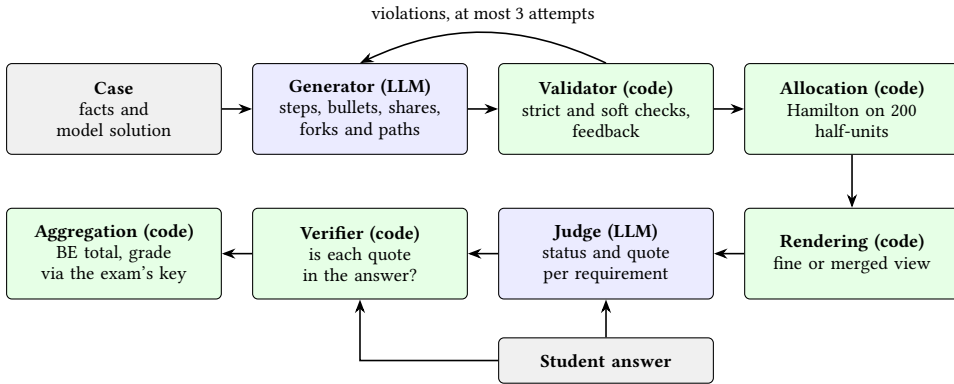


Fig. 1. The grading-sheet pipeline. Blue: model steps; green: code. The judge also receives the case facts and the model solution as context. Levers per stage are listed in Table 2.

4.2 Generation: the checklist sheet

The generator receives the case text, the model solution and a fixed output specification, the same for every model, through plain text completion. It returns one JSON document:

- *Sections and steps.* Each step names one expected legal performance and carries a relative weight. Klausurschwerpunkte (the exam’s focal problems) are marked groups of steps.
- *Requirement bullets.* Each step has 1 to 6 bullets, each with a relative share and a standard (*Maßstab*): *richtig* (judged against the correct law) or *folgerichtig* (judged against the candidate’s own earlier decision, the consequential-error principle at bullet level). A bullet describes a performance that stands in the answer and can be quoted. No bullet may be met by an absence, none may be conditional, and none may ask only for a norm, a term or a result to be named. A step above 3 BE needs at least two bullets.
- *Forks and solution paths.* A dispute that changes only a result or its reasons stays inside a step as an outcome-neutral bullet. A structural fork (*Weichenstellung*) exists only where another tenable decision drops a step of the model solution’s path or adds a step with its own prerequisites. Each alternative path (*Lösungsweg*) carries its own steps, the reasoning that makes it tenable (*Vertretbarkeitsanforderungen*), and the budget of the steps it replaces.
- *Owed auxiliary opinions.* Steps or sections can be marked as owed even when the candidate’s own decision cuts them off (*Hilfsgutachten*), for instance when a contractual claim already fails at formation. The marker names the triggering decision.
- *Work results.* Each step belongs to an *Arbeitsergebnis*, such as an opinion or a draft judgment.

The model states weights, never points. How it states them is a lever with three settings: flat weights over all steps; hierarchical shares, where each level states its share of the parent (Roth’s paradigm, Z2); and top-down BE, where the model writes section totals first and then the steps inside (the D2 paradigm), normalized by code.

4.3 Allocation by code

Code turns weights into BE. It apportions 200 half-units (100 BE) over the steps by the largest-remainder (Hamilton) method [3], with a floor of 0.5 BE per step. The floor is enforced before the remainders are ranked: steps whose quota falls below the floor are pinned to it, and the rest is apportioned again. Every free step then lands within one half-unit of its quota among the free steps, a heavier step never receives less than a lighter one, and weights that already sum to the

total pass unchanged. A bullet is worth the step's BE times the bullet's share. Budgets of alternative paths equal the sum of the steps they replace. A step above 10 BE fails validation with a message that asks the model to split the step or lower its weight.

The first iteration's routine floored first and then ranked the remainders of the unfloored quotas. An audit of its 170 sheets found 27 steps in 11 sheets outside their fair range and 12 order inversions in 3 sheets; 1 of the 15 sheets used in the main arm was affected (Appendix C). The corrected routine removes both defects. On the half-point grid it also keeps distinct weights apart: the as-run routine gave equal points to 458 pairs of steps with different weights, the half-point routine to 0.

4.4 Validation and retries

A deterministic validator checks every document in two tiers (Appendix I). A strict violation fails the attempt. The model then receives its previous document and the verbatim violation messages, for at most 3 attempts. Strict checks cover the structure and all cross-references, absence and conditional bullets, steps named as a result, big steps with a single bullet, forks that replace a whole section, and paths that repeat the steps they replace. Soft checks only log hints, for example when most steps exceed 4 BE. Every sheet records content hashes of the validator and the prompt. A lexical check cannot see paraphrase, so two steps that say the same thing in different words pass.

4.5 Rendering

Code renders the sheet for the judge: sections, steps with their BE, bullets with their share in percent and their standard, forks with their paths and tenability requirements, and notes on owed auxiliary opinions. A second view merges consecutive steps of a section until a merged step reaches about 10 BE, without crossing a focal group, a fork or an auxiliary-opinion marker. Totals do not change. The two views isolate step size as a lever.

4.6 Judging

The judge receives four tagged inputs with fixed roles: the case, the model solution (a reference, not the answer), the rendered sheet and the answer. It returns JSON under a schema derived from the sheet. There are 3 scoring units. In the *bullet* unit the judge marks each requirement as met, partly met or not met, which code credits with 1, 0.5 or 0 of the bullet's BE. In the *step* unit it states half-point BE per step, as on a classic sheet. In the *rating* unit it grades each step from 0 to 18 against the statutory band definitions. Every credit needs a verbatim quote from the answer. The judge's rules are fixed:

- *Placement*. Work counts for the step whose requirement it meets, even if it stands elsewhere in the same question and work result; the step is then flagged as misplaced. It counts once, and work on another question counts only if the answer refers to it explicitly.
- *Consequential errors*. *Folgerichtig* bullets are judged on the answer's own earlier decisions. Other bullets are judged on the correct law, but an error that has already cost credit is not charged again; the judge records the chain.
- *Auxiliary opinions*. Work on questions that the answer's own decision cut off earns credit like work in the main opinion. It never replaces the main result.
- *Alternatives*. In design A0 (*replace*) the judge puts the work on another tenable path onto the replaced steps. In the *branch* designs the judge scores every path's steps and declares, with a quote, which path the answer follows; a path counts as followed only if the answer argues it as its tenability requirements demand. A2 counts the declared path (the model solution's path if the quote fails), A1 the best-scoring path; one call yields both.

Table 2. Levers of the pipeline. Baselines: the first-iteration step-score sheet, the holistic instrument, and the exam author’s own sheet graded by the same judge.

Stage	Fixed in code	Levers
Generation	specification, validator, retries	generator; allocation (flat, hierarchical, top-down)
Rendering	layout, shares in percent	view (fine, merged)
Judging	rules, schema, quote check	judge; unit (bullet, step, rating); alternatives (A0, A1, A2)
Aggregation	rounding, grade key	passes averaged (1–3); two-judge mean

- *Unforeseen paths.* If the answer takes a tenable path that the sheet does not foresee, the judge scores the functionally corresponding step at equal depth of reasoning and flags it; such a script goes to human review.
- *General rules.* Contradictory results without a decision fail the requirement on the result. Unnecessary alternative analyses earn nothing. Length, frequent citations and strings of keywords are not performance, and a short, correct application meets a requirement fully.

Instructions inside the tagged inputs are ignored. A diagnosis block records the assessment status (scored, review required, not evaluable), the work results, owed auxiliary opinions, error chains and short hints for the candidate; it never changes points.

4.7 Evidence verification and its limits

Code checks every quote against the answer (Appendix H). It normalizes layout and compares tokens in order, with small tolerances: up to 2 skipped answer words between quoted words, one missed quoted word per 8, and an inflection or a one-letter typo on longer words. *Protected* tokens must match exactly and may never be skipped: negations, result words and their antonyms, qualifiers, Roman numerals and every token with a digit. A match fails when a negation or a condition stands right before it in the same clause, or right after it in the same sentence. A fragment needs at least 3 tokens and more than a norm citation. An unverified quote sets the item to zero; the model’s own value stays in the output for analysis.

The check has limits. It tests presence, not relevance: a real sentence can be quoted for the wrong requirement, and only the judge decides what a sentence proves. Correct work that the judge paraphrases instead of quoting earns nothing. A sentence that the answer copied from the case text verifies. The edge rules also reject some correct quotes, for instance one cut before a trailing condition that does not change the result. The first iteration had no verifier, and the platform grades that the D2b students saw ran with an earlier version. Every arm reports its verifier version. [PENDING: regression replays at the freeze: quotes re-checked on D1, D2 and the stored D2b grades, and how many flip]

4.8 Aggregation and grades

A step’s points are the sum of bullet credits times bullet BE (bullet unit), the stated BE (step unit), or the BE equivalent of the step grade (rating unit): the midpoint of the grade’s band under the exam’s key, with 18 points worth the full step. Code sums the unrounded points and rounds once, half up, to the half-point grid; the step scores shown are distributed from that total by largest remainder so that they add up to it. The grade follows from the total via the exam’s key. For the rating unit we also report the weighted mean of the step grades, Roth’s aggregation. Passes can be averaged.

5 Data and design

5.1 D1: the first-iteration corpus

D1 holds the 15 exams of the BenGER corpus [34], from civil, public and criminal law, with 45 answers: per exam one traditional, one co-created and one model-generated solution. The 30 human-written answers each have grades from 3 blind raters and the answer’s creator; agreement uses the mean of these 4 grades. The blind raters agree at $ICC(2,1) = .63$ and differ by 23.9 raw points per answer on average. The prior study’s primary judge tracked the pool at $r = .76$ with an MAE of 10.8 raw points, or 2.7 grade points when the MAE is computed on the grade scale. D1 is our only data across many exams, and its human reference is noisy.

5.2 D2: the expert exam

D2 is one police-law exam from LMU, set in yearly runs from 2024 to 2026. Its author is a co-author of this paper. His grading sheet has 46 steps and 100 BE on a half-point grid, and his key passes a script at 40%. The sheet exists in 3 states with the same steps: the 2024 run used earlier step maxima, and 9 scripts of 2025 carry older step labels; the other 2025 scripts and the 2026 run use the current sheet.

- *D2a* holds 15 typed scripts from the runs of 2024 (3) and 2025 (12), each with per-step grades on the sheet state it was graded on. We call these the *reference grades*: for the 9 scripts with older labels the grader is still being confirmed. The judge sees the current sheet for all scripts.
- *D2b* holds 25 submissions from the 2026 run on our platform. We exclude 2 uploads from the author’s own account (a test upload and a copy of a *D2a* script), leaving 23 student submissions. The platform graded them at submission time with GPT-5.4 Mini, and the students saw these grades. The author grades the 23 submissions blind, without access to the model grades. [PENDING: blind expert grades for D2b]

In total D2 has at most 38 distinct scripts, 37 if one pair with overlapping text counts once. The case text changed between runs, at least in its dates. 12 *D2a* scripts are judged against their own year’s text; the 3 scripts of 2024 are judged against a later text and flagged. [PENDING: the 2024 case text from the exam author] A second grader will grade 20 scripts blind (10 from each part) to give a human-to-human ceiling. [PENDING: second grader]

Before any research call to a model provider, all D2 texts are pseudonymized and checked for personal data. The check found and removed 1 line with a student’s name and 1 examiner header; 9 grade spreadsheets carried a private e-mail address, which no derived file contains. The expert’s sheet never reaches a generator or a holistic judge. A canary check asserts that no prompt of any other arm contains terms of the expert sheet or of the D2 solution.

We also hold Gregor Roth’s sheet and grading workbook for a civil-law exam (Z2), without case, solution or scripts, and use them only for their scoring format (E7).

5.3 Experiments

The development data are D1 and D2a. D2b is held out and used once, after the author has graded it, to confirm the pipeline chosen on the development data. Table 3 lists the experiments. Each runs on its own budget and only after the owner’s approval.

Measures. Against the expert we report the mean absolute error (MAE) and the signed bias of the total, $ICC(A,1)$ [31], the share of grades within one grade point, and pass/fail agreement (Cohen’s κ); on D1 also the concordance correlation [29] with the human pool. Stability is the SD across repeated passes on the same sheet (*repeat SD*) and across independently generated sheets for the same exam (*sheet-sample SD*). Robustness is the probe battery. Faithfulness is the share of quotes

Table 3. Experiments of the extended study, with planned costs in USD at catalog prices and pilot-measured token counts. [PENDING: actual costs per experiment]

Experiment	RQ	Scope	USD
E0 Gate v2 and instrument freeze	RQ1, RQ5	D2 and 4 D1 exams; the expert sheet (step and rating units), one checklist sheet each from two generators, the first-iteration sheet; 3 judges	15
E1 Generator benchmark	RQ1, RQ3, RQ4	12 generators, 15 D1 exams and D2, 2-3 samples per exam, one reference judge (2 passes), a second judge on 4 generators	170
E2 Judge benchmark	RQ1, RQ2, RQ5	6-8 judges on fixed sheets; 45 D1 answers and 15 D2a scripts, 3 passes	100–150
E3 Generator x judge interaction	RQ1, RQ4, RQ5	3 generators x 3 judges x 15 D2a scripts x 2 sheet samples x 2 passes, plus 5 D1 exams	25
E4 Pipeline optimization	RQ6	fractional factorial over the levers on D1 and D2a; one confirmation on held-out D2b	80
E5 Alternatives, Hilfsgutachten, placement	RQ2, RQ6	A0 vs A1 vs A2 on scripts that take another path; alternative-path and misplacement probes	15
E6 Grading practice	RQ7	re-aggregation of stored outputs (BE distribution, method bullets, pass mark sweep, key shape); Kardinalfehler probes	10
E7 Comparison with Roth’s paradigm	RQ7	scoring format and weight elicitation on D2 (Z2 has no case or scripts); re-aggregation	15
Total			430–480

the verifier rejects and the gap between the model’s total and the verified total. Cost and latency are measured per call.

Probe battery. Before any benchmark run, each judge and instrument passes a battery of synthetic answers over 3 passes (Appendix G): negative probes (an empty answer, the case text, another exam’s solution, a keyword list, an injected instruction) must score near zero; the exam’s own model solution must score high; removing one section of the model solution must cost about that section’s BE; moving correct reasoning under another heading must cost almost nothing.

Estimation. We estimate rather than test. Intervals are bootstrap CIs clustered by exam (D1) or by script (D2); variance components come from mixed models; generator and judge ranks carry bootstrap intervals. The data limit what we can see (Appendix F). With 15 D2a scripts, a paired difference in absolute error below about 10.9 points is not detectable (power .8, $\alpha = .025$, t -based); with all 38 scripts the bound is about 6.4 points. On D1’s 45 answers a change in repeat SD below about 0.88 is not detectable. Each judge runs at the platform’s grading default temperature; we report each provider’s determinism. GPT-5.6 Luna cannot be seeded, so its repeat SD mixes sampling noise with instrument effects.

Development history. Before the benchmark we ran small development runs to build the instrument: a feasibility pilot, two probe gates and two generation runs on D2. They used an earlier template, had gaps in the verifier, and one probe was defective. Together with 4 independent review rounds they shaped the instrument; their numbers are not evidence. They cost USD 11.47.

6 Results

6.1 First iteration

The first iteration asked 12 generators for one step-score sheet per exam. Each step carried a weight class, ordered partial-credit levels, provenance and consequential-error rules; alternative paths

Table 4. First iteration: stability and agreement by instrument. SD_{rep} : repeat SD over 3 passes on 45 answers. r , MAE, bias: first pass against the human pool on 30 human-written answers; bias is judge minus humans. Step-score totals are step sums. [†]Stored total; the step-sum recompute needs withheld run outputs [PENDING: few-shot rows and zero-shot bias on step sums]. The 6-judge panel SD ranges from 6.98 to 10.23 across the panel definitions.

Instrument	SD_{rep}	r	MAE	bias
<i>Judge: GPT-5.6 Luna</i>				
Holistic	2.25	.80	11.2	+4.3
Step-score sheet, zero-shot	1.96	.77	12.8	-6.9 [†]
Step-score sheet, few-shot	1.80 [†]	.70 [†]	15.8 [†]	-7.9 [†]
<i>Judge: GPT-5.4 Mini</i>				
Holistic	3.87	.75	11.2	-2.7
Step-score sheet, zero-shot	5.51	.41	19.4	+7.4 [†]
Step-score sheet, few-shot	5.22 [†]	.36 [†]	21.3 [†]	+5.8 [†]

replaced named steps within their budget. A seeded random draw fixed one valid sheet per exam for all judges. Judges graded the 45 answers against the drawn sheet and, in matched arms, against the *holistic* instrument of the prior study [34] (the same 10 dimensions for every exam) over 3 passes. A few-shot arm regenerated sheets with one exemplar from another exam. We report totals as step sums: the first-iteration pipeline sometimes stored the model’s own total when it lay within half a point of the step sum (Appendix A).

Generation (RQ3). 170 of 180 generator–exam cells hold a valid sheet. 10 generators covered every exam; Llama-4 Maverick covered 12 and GPT-5.4 Mini 8, mostly because feedback did not repair broken references of alternative paths. Of 220 dispatched series, 50 failed: 34 aborted at the provider (all GPT-5.4 Mini, credit errors), 14 failed validation and 2 failed parsing. First-attempt compliance over all series that reached the model spans 13% to 100%. Counted over valid sheets only, GPT-5.4 Mini looks better than it is: 38% there, but 17% over the series that reached the model (3 of 18) and 6% over all 52 of its series (Table 8). Generators did not converge: mean step counts range from 6.3 to 25.5, and stated weight sums from 34 to 179. A lexical screen found vague formulas that the prompt itself bans in 7 of 170 sheets (16 with near variants). The generation prompt asked for “exactly 100 raw points” in its first line, while its weighting rule said that no exact sum was needed because code normalizes; we disclose this as an ambiguity. The weight class was shown to the judge but played no part in the arithmetic.

Variance (RQ1). For Luna, repeat SD falls from 2.25 (holistic) to 1.96 (zero-shot sheet) and 1.80 [PENDING: few-shot on step sums] (few-shot sheet; Table 4). The direction is consistent, but no single step is established: few-shot minus holistic is -0.45 (95% CI -1.01 to +0.09; sign test $p = .45$), zero-shot minus holistic -0.28 (95% CI -0.76 to +0.16; $p = .65$), and few-shot minus zero-shot -0.17 (95% CI -0.64 to +0.25; $p = 1.00$). Across the 5 judges with matched arms, the zero-shot effect averages +0.03 (95% CI -0.62 to +0.73; Table 10). This is the unweighted mean of the per-judge differences, without correction for multiple comparisons. Only Sonnet’s per-judge interval excludes zero, and it points the adverse way: Claude Sonnet 4.6 is steadier with the holistic instrument (+0.36, 95% CI +0.10 to +0.65). Below the capability floor the sheet destabilizes the judge: GPT-5.4 Mini’s repeat SD rises from 3.87 to 5.51 (+1.65, 95% CI -0.17 to +3.46, clustered by exam; cell-level sign test $p = .04$, Wilcoxon $p = .008$). The steadiest few-shot sheets, a post-hoc subset by generator, reached 1.26 with Luna, and an in-sample floor (the best sheet per exam, chosen after the fact) 1.01;

both are optimistic bounds without intervals. These contrasts use stored totals; step sums move the zero-shot means of the matched judges by at most 0.02.

Agreement (RQ2). Agreement with the human pool fell as the sheets tightened. Luna's r moved from .80 to .77 and .70, and its bias flipped from +4.3 to -6.9 and -7.9; the few-shot exemplars had been chosen for stability alone. Its zero-shot MAE rose by +1.6 (95% CI -3.3 to +7.0). GPT-5.4 Mini graded adequately with the holistic instrument ($r = .75$, MAE 11.2) but not with the sheet: $r = .41$, MAE 19.4, an increase of +8.2 (95% CI +2.0 to +15.8). Few-shot sheets left it there ($r = .36$, MAE 21.3). Leave-one-exam-out bias correction shows that the error of the prior study's control judge was mostly a level shift (MAE 14.9 to 9.5), whereas Luna's zero-shot error was not (12.8 to 13.1).

Judges and probes. The 7 roster judges graded all answers against the drawn sheets (Table 9). Apart from Mini they reached r .70 to .77; Claude Opus 5 and GPT-5.6 Terra, one pass each, .74 and .75. A battery of 45 degenerate answers per type (the case text verbatim, a near-empty answer, another exam's model solution) and each exam's own model solution as a positive control tested execution (Table 12). Only Claude Opus 4.7 and DeepSeek-V4-Pro passed the negative battery on every pass. Luna credited the near-empty answer on 26 of 45 zero-shot passes and 24 of 45 few-shot passes, with 96.2 and 97.1 points on average when it did [PENDING: median over failing passes], but on 0 holistic passes. GPT-5.4 Mini failed 100 of 135 negative passes under the sheet and 125 under the holistic instrument, but its holistic failures were small (median 5 points on the empty answer) and its sheet failures near-total (median 95). Under the sheet, the off-topic answer scored at most 4.5 points with every judge except Mini (up to 94). All 7 judges passed the positive control under the sheet; Mini failed it holistically on 13 of 45 passes. On one answer that only restates the case, Mini under the sheet gave 99, 100, 100 points and Luna 0, 0, 0.

These failures belong to the first-iteration harness. It had no quote verification and a generic system prompt. Its response schema listed the steps in storage order (by key length, then bytes), not in outline order, and the steps of alternative paths had no response keys, so a judge had to move credit for another path onto the replaced steps, where the per-step cap dropped any overflow [PENDING: number of alternative paths affected]. Budget neutrality thus held in the document, not in scoring. The new pipeline changes all of this at once. E0 and E2 must therefore measure the capability floor again before we claim anything about it, and E4 can separate the levers only in part.

Generators and the audit (RQ4). A crossed sweep graded each exam's answers under every valid sheet of that exam: 170 sheets and 510 sheet-answer cells, Luna, 3 passes (Table 11). An automated audit by GPT-5.4 Mini rated each blinded sheet on 7 dimensions from 1 to 5. Within exams, the audit total shows no linear association with grading error ($r = .09$, 95% CI -0.08 to +0.26; Spearman .09) and a weak one with stability ($r = -.13$, 95% CI -0.33 to -0.09; Spearman -.22). It correlates with the *absolute* bias ($r = .19$, 95% CI +0.00 to +0.35) [PENDING: within-exam correlation with signed bias]. At the generator level ($n = 12$), higher audit totals go with harsher grading (Spearman -.70 with Luna, -.63 with Sonnet): the Claude sheets top the audit (29.1 and 29.0 of 35) and grade among the harshest (bias -11.0 and -14.5). DeepSeek-V4-Flash sheets cut agreement ($r = .40$, the lowest) behind a mid-field audit (27.0). Llama-4 Maverick sheets made Luna least stable (repeat SD 4.07), but the audit ranked them lowest (22.3), so this failure was visible. GPT-5.4 Mini, the weakest judge and least compliant generator, wrote the best sheets when it wrote any (paired Δ MAE -3.2 on 8 exams), but its probability of ranking first is .47 and the estimate is conditional on survival. The ranking replicates moderately under a Sonnet lens (Spearman .50; .86 among full-coverage generators). Few-shot generation raised Mini's valid sheets from 8 to 10 of 15; since the few-shot arm also used its own pool of 5 generators, its effects are confounded with that pool.

Weights and step size. The generated weights carry a small, judge-dependent signal (Appendix D). Re-aggregating the stored per-step scores, Luna keeps most of its agreement with class weights

Table 5. Probe battery on the expert exam. Battery: six degenerate answers (empty, repeated case text, off-topic, same-area off-topic, keyword list, injected instruction) must score 0 or near 0 on every pass; max is the highest score any of them received. Proposed: the highest of the judge’s per-probe median totals on those answers before the evidence rules. ML: the model solution. Ablation: points lost when one part of the model solution is removed, observed / expected; the expected loss is the sheet’s BE for the steps matched by name to the removed part, a heuristic. Misplaced: points lost when one correct paragraph is moved under another heading. Flip: points lost when the results of the model solution are negated.

Instrument	Judge	Passes	Battery	Max	Proposed	ML	Ablation	Misplaced	Flip
Expert sheet (step unit)	Luna	2	pass	0.0	23.0	99.5	19.5 / 21.0	1.3	18.3
Expert sheet (step unit)	DeepSeek	1	pass	0.0	0.0	99.5	20.5 / 21.0	0.0	18.0
GPT-5.4 checklist sheet	Luna	2	pass	0.0	0.0	96.0	23.5 / 25.0	0.5	10.0
GPT-5.4 checklist sheet	DeepSeek	1	pass	0.0	0.0	94.5	21.0 / 25.0	0.0	5.0
GPT-5.4 Mini checklist sheet 1	Luna	2	pass	0.0	0.0	98.5	23.3 / 26.5	1.8	11.3
GPT-5.4 Mini checklist sheet 1	DeepSeek	1	pass	0.0	0.0	100.0	26.0 / 26.5	0.0	15.0

alone (r .764 vs .769) and loses some with uniform weights (.724). For DeepSeek-V4-Pro, class weights do worse than uniform ones (.683 vs .691). Uniform weights lowered repeat SD for 6 of 7 judges, and the actual weights beat 75 to 99% of within-sheet permutations. The judges saw the actual step maxima, so these re-aggregations are conditional on them. Step size matters more than step count (Appendix E). Steps of 15+ points held 22% of the points but a larger share of the summed per-step variance for 5 judges: Luna 36%, Gemini 43%, Sonnet 32.5%, DeepSeek-V4-Pro 65%, and Opus 4.7 41% from 15% of its points (21 step cells). For Qwen (25%) and Mini (28%) the share was close to proportional. Across the 170 sheets, step count correlated only weakly with error (r = .14) and with repeat SD (–.14). The new instrument therefore uses small steps and tests step size as a lever.

6.2 Preliminary results on the expert exam

These runs are a first look at the expert exam, not the benchmark: the pipeline as it stood (output specification checklist-3), sheets from two generators, and up to three judges. They include no run without a sheet, so they speak to validity (RQ2) and generation (RQ3), not yet to variance reduction (RQ1). The reference grades of 15 D2a scripts serve as the yardstick; the exam author has yet to confirm who graded 9 of them, so we call them reference grades, not expert grades.

Generation. GPT-5.4 produced a valid sheet in 1 of 4 chains (34 steps), GPT-5.4 Mini in 2 of 4 (19 and 22 steps; the validator flagged sheet 1 as coarse). Of the 5 failed chains, 3 ended on the 10-BE cap, 1 on the rules for forks and 1 on an unreadable response. GPT-5.4 builds richer alternative paths, and 6 of its 10 failed attempts broke the rules on forks. For GPT-5.4 Mini, 3 of 6 failed attempts exceeded the cap: a generator states relative weights, not BE, and on a solution path only splitting a step helps, which the feedback did not say. A valid sheet cost USD 4.15 with GPT-5.4 and USD 0.23 with GPT-5.4 Mini, failed chains included.

Probe battery. Luna (2 passes) and DeepSeek (1; its second pass stopped for time) passed the battery on every instrument they were gated on (Table 5): no degenerate answer received points. The plan asks for 3 passes, and GPT-5.4 Mini as a judge and the second GPT-5.4 Mini sheet were not gated; their rows in Table 6 are marked. Removing one part of the model solution cost about as many points as the sheet assigns to that part, and moving a correct paragraph under another heading cost almost nothing. On the expert sheet in the step unit, Luna still proposed half a point for every step of an empty or off-topic answer, and only the rule that a credit needs a quote removed them; on the checklist sheets it proposed nothing in the first place.

Table 6. Totals on the 15 D2a scripts against the reference grades (grader confirmation pending for 9 scripts; 3 scripts of the earliest cohort are judged against a later case text). Each statistic is computed per pass and averaged over the passes; MAE with a script-bootstrap 95% interval; bias is judge minus reference; pass/fail agreement and Cohen’s κ at the exam’s pass mark of 40 BE; SD_{rep} : mean SD over repeated passes. [†] not gated by the probe battery.

Instrument	Judge	Passes	MAE [95% CI]	Bias	r	Agr.	κ	SD_{rep}
Expert sheet (step unit)	Luna	2	4.6 [3.3, 6.0]	−1.9	.95	80%	.53	1.7
Expert sheet (step unit)	DeepSeek	1	5.2 [3.2, 7.4]	−3.9	.95	93%	.86	–
Expert sheet (step unit)	Mini [†]	1	9.4 [6.4, 12.7]	−1.6	.74	67%	.21	–
GPT-5.4 checklist sheet	Luna	2	6.5 [4.3, 9.1]	+3.9	.90	87%	.67	2.3
GPT-5.4 checklist sheet	DeepSeek	1	6.2 [4.1, 8.5]	+3.1	.91	80%	.47	–
GPT-5.4 checklist sheet	Mini [†]	1	10.3 [6.6, 14.2]	+5.7	.74	67%	.12	–
GPT-5.4 Mini checklist sheet 1	Luna	2	8.6 [6.0, 11.5]	+8.1	.93	77%	.36	2.8
GPT-5.4 Mini checklist sheet 2	Luna [†]	2	8.2 [5.6, 11.1]	+8.2	.94	73%	.25	1.8

Agreement with the reference grades. On the expert’s own sheet, a single Luna pass came within 4.6 points of the reference grades on average (bias −1.9, $r = .95$) and agreed on pass or fail for 80% of the scripts ($\kappa = .53$); two passes differed by 2.3 points on average (Table 6). GPT-5.4 Mini, the platform’s current default judge, was further off on the same sheet (MAE 9.4, $r = .74$); paired over the scripts, its absolute error was +4.8 [1.6, 8.3] points higher. DeepSeek-V4-Pro reached an MAE of 5.2 on the expert sheet and 6.2 on the GPT-5.4 sheet. With generated sheets, Luna’s MAE was 6.5 (GPT-5.4 sheet) and 8.6 (GPT-5.4 Mini sheet 1). The change is mostly leniency: the signed error rose by +5.8 [3.0, 8.8] and +10.1 [7.4, 12.7] points, the absolute error by +1.9 [−0.2, 4.2] and +4.0 [1.4, 6.6], while the order of the scripts stayed close to the reference ($r = .90$ and $.93$). Two readings follow, both preliminary. First, the judge mattered at least as much as the sheet: the default judge on the expert’s sheet was not closer to the reference than Luna on the GPT-5.4 sheet (difference +2.9 [−1.3, 6.9]). Second, the generated sheets were more lenient than the expert’s; a generated sheet needs a stricter specification or a calibration step before its totals can stand in for an examiner’s. A second GPT-5.4 Mini sheet from the same input gave Luna an MAE of 8.2 (bias +8.2). A script’s total differed between the two sheets by 3.5 points on average. Separating the pass noise, the sheet-to-sheet SD is 2.1 and the pass-to-pass SD 2.7: the sheet a generator happens to write adds variance of the same order as the judge’s own noise, so a benchmark has to sample several sheets per generator.

These numbers come from 15 scripts of one exam. The observed spread of the paired differences puts the smallest detectable difference in absolute error at 2.5 to 7.7 points (80% power, $\alpha = .05$); we read them as a first signal for the benchmark design, not as a ranking.

6.3 Benchmark results (E1 to E7)

[PENDING: E1 to E7]. The generator benchmark (E1) will report compliance, validator defects, the author’s spot audit of 10 bullets per sheet, validity on D1 and D2a, repeat and sheet-sample SD, alignment with the expert sheet, and the cost per valid sheet. The judge benchmark (E2) will report validity against the expert (MAE, bias, ICC(A,1), grades within one point, pass/fail κ), per-step agreement, repeat SD, the probe battery, the share of rejected quotes and cost. E3 will estimate variance components and test whether generator rankings hold across judges (Kendall’s τ). E4 will report the Pareto front of error, variance and cost and confirm the recommended configuration once on the blind expert grades of D2b. E5 compares the designs for alternative paths; E6 and E7 answer the questions of Section 2 and compare the two expert paradigms on D2.

7 Discussion

[DRAFT: the answers per research question are written after E1 to E7]

What the first iteration already shows. First, a generated sheet does not reduce variance by itself. Whether it steadies a judge depends on the judge and the sheet together: across the capable judges the average effect was null, and one judge was steadier without a sheet. Second, mean agreement hides a failure that matters for every grade a student sees. Under detailed sheets, some judges gave near-full credit to empty and copied answers; Luna credited the empty answer on 26 of 45 passes under the sheet and on 0 with the holistic instrument. A probe battery must therefore gate every judge and instrument before use, and code must tie every credit to the answer. Third, writing, using and auditing a sheet are different skills. The least compliant generator wrote the best sheets when it succeeded, and the automated audit favoured sheets that graded harshly. Generator and judge must be chosen together, by outcome.

Answers to the research questions. *[PENDING: one paragraph per research question after E1 to E7]*

Implications for practice. The pipeline is a research instrument. Its sheets make each credit auditable: an examiner can see which requirement a quote met and can correct the sheet itself, which a holistic score does not allow. Two uses fit the evidence so far. One is formative feedback on practice exams, where students already receive LLM grades and quote-backed credit makes that feedback checkable. The other is support for examiners who write sheets, where a generated draft is edited, not trusted. Grade-bearing assessment needs a human examiner. *[PENDING: recommended pipeline and its cost per script]*

8 Limitations

One expert exam. D2 is one case in one area of law, graded by its author. D1 spans many exams, but its human reference is noisy. Results on D2 may not transfer to other exams, areas or examiners, and the second grader gives a human ceiling for one exam only.

Open data questions. For 9 D2a scripts the grader is not yet confirmed, and the 3 scripts of 2024 are judged against a later year's case text. The judge sees the current sheet also for scripts graded on an earlier state. The D2b students saw LLM grades before the author grades their scripts; he grades without seeing them.

The exam author is a co-author. He wrote the exam, grades the scripts, audits generated sheets and co-writes this paper, so he knows the study's aims. Blind grading of D2b and the second grader limit this, but his spot audit of generated sheets is not independent.

Leakage. During development, examples in the generator and judge prompts came from the D2 exam. We replaced them with neutral examples and added a canary check. The check is lexical and cannot rule out that the prompts were shaped while the expert sheet was in hand. The step-size guidance comes from the expert sheets by design.

Verification. The verifier checks that a quote exists, not that it supports the credit (Section 4.7). It rejects some correct quotes and cannot catch a real sentence quoted for the wrong requirement.

Models and sampling. GPT-5.6 Luna cannot be seeded. Even at temperature 0, DeepSeek-V4-Pro's totals varied between passes (repeat SD 1.81 in the first iteration), so repeat SD mixes model, sampling and serving effects. Providers change and retire models; we record model ids and dates. The prior study's repeat baseline ran on a judge that has since been retired; the matched arms remove this confound from every contrast we interpret.

Exploratory design. Nothing was pre-registered, many comparisons are reported, and the precision limits of Section 5.3 apply. The held-out confirmation on D2b is used once. The first-iteration results are exploratory too; several of its arms were added after earlier results were known.

9 Ethics

Consent and personal data. The students whose D2 scripts we use consented to research use, for the 15 scripts of D2a and the 25 platform submissions of D2b. The consent does not extend to sending their personal data to model providers. All D2 texts are therefore pseudonymized and checked for personal data before any research call (Section 5.2). The platform grades that the D2b students saw were produced in the normal use of the platform, before this study. [PENDING: owner to confirm the wording on the platform-side grading and the providers' data terms] D1 reuses the prior study's corpus under the consent obtained there, with pseudonymized graders [34].

Roles. The exam author is a co-author, the source of the reference grades and the auditor of generated sheets. We report this as a limitation and mitigate it with blind grading and a second grader.

Students who saw LLM grades. The D2b students received LLM grades as feedback on a practice exam. [PENDING: owner: whether and how students learn the expert grades]

Use of the instrument. Generated sheets are studied as a research instrument for evaluating and supporting grading. They are not a replacement for human examiners in grade-bearing assessment, where legal and pedagogical safeguards apply.

Roth's sheet. Gregor Roth agreed to be named and to have his sheet and workbook used. We report only their structure, not their content.

10 Data and code availability

The platform, including the judge, the verifier and the aggregation code, is open source at <https://github.com/SebastianNagl/benger-platform>. Its folder publications/Transformation holds the analysis scripts and every aggregate this paper reports (make_analyze-cslaw; the D2 and Z2 aggregates need the private data, make_analyze-d2). The generator is part of a proprietary extension of the platform. [PENDING: owner decision: release of the generator prompt, output specification and validator rules] For D1 we follow the first study: its exam texts are third-party material and are not redistributed, and the generated sheets are withheld with them because they enumerate the model solutions. D2 and Roth's sheet are private, and we publish only aggregates (Appendix L).

11 AI usage

For the first iteration the authors used OpenAI GPT-5.6 Sol to discuss approaches and Anthropic Claude Fable 5 for coding and writing assistance. For the extended study, Anthropic Claude Opus 5.5 was used for coding, for 4 independent review rounds of the instrument and the analysis, which are part of the method record, and for writing assistance, including an AI-assisted literature scan. All results were verified by the authors, and all citations were checked against their sources. The authors take full responsibility for the paper.

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Table 7. Defects of the first-iteration pipeline and how the rebuilt pipeline handles them.

Topic	Defect and handling
Apportionment	The routine broke the quota rule for 27 steps in 11 sheets, with 12 order inversions in 3. Of the 15 drawn sheets, 1 is affected (DeepSeek-V4-Pro); all 11 affected sheets enter the generator ranking. Now floor-constrained Hamilton (Section 4.3).
Totals	The stored total was sometimes the model’s own, within half a point of the step sum. With step sums the zero-shot repeat SD moves for Luna from 1.97 to 1.96, for Mini from 5.52 to 5.51. Now code computes every total. [PENDING: few-shot rows]
Judge schema	Response keys followed storage order, not outline order. Now outline order (Section 4.6).
Alternatives	Steps of alternative paths had no response keys, and points above a step’s cap were dropped. Now every path has its own keys (Section 4.6). [PENDING: count of affected paths]
Verifier	The first iteration verified no quotes; the stored D2b grades used an earlier verifier.
Prompt wording	The generator prompt asked for exactly 100 points and also for relative weights; the weight class was rendered to the judge but unused. Now the model states weights only and code allocates.
Few-shot pool	The few-shot arm changed the generator pool along with the exemplar, so the two effects are confounded.

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A Defects of the first-iteration pipeline

The first iteration’s pipeline had the defects below. The rebuilt pipeline removes each of them. Where a defect touches a first-iteration number, this paper reports the number without it, for instance totals as step sums (Table 7). The numbers come from the file `first_iteration.json` in `data/processed/cslaw`, which the script `cslaw_first_iteration.py` in `scripts/analysis` builds.

B First-iteration tables

Tables 8 to 12 give the first iteration’s per-generator, per-judge and per-probe results that Section 6.1 summarizes.

Table 8. First iteration, generation by generator: valid sheets, first-attempt compliance over successful sheets and over all series that reached the model, mean attempts per valid sheet, mean step count, and automated-audit total (of 35).

Generator	Valid	1st (valid)	1st (series)	Attempts	Steps	Audit
Claude Opus 4.7	15/15	73%	73%	1.27	17.1	29.1
Claude Sonnet 4.6	15/15	13%	13%	1.93	22.0	29.0
DeepSeek-V4-Flash	15/15	80%	71%	1.27	18.5	27.0
DeepSeek-V4-Pro	15/15	100%	100%	1.00	25.5	26.6
GPT-5.4	15/15	100%	100%	1.00	16.8	28.5
Gemini 3.1 Flash-Lite	15/15	80%	80%	1.20	6.3	24.4
Gemini 3.1 Pro	15/15	100%	100%	1.00	15.3	26.5
Qwen3-235B-Thinking	15/15	40%	40%	1.60	12.8	25.6
Qwen3.5-122B	15/15	13%	13%	1.93	9.3	26.0
Qwen3.6-35B	15/15	60%	56%	1.60	11.5	26.3
Llama-4 Maverick	12/15	75%	60%	1.25	6.5	22.3
GPT-5.4 Mini	8/15	38%	17%	1.88	17.1	27.4

Table 9. First iteration, judges on the drawn step-score sheets (step sums), sorted by MAE. r and MAE: first pass against the human pool (30 answers). SD_{rep} over 3 passes on the listed number of answers. Battery: fail(N) counts failing negative passes. P01: the first pass on an answer that only restates the case. Below the line: single-pass judges, from the database extraction.

Judge	r	MAE	SD_{rep}	Answers	Battery	Pos.	P01
GPT-5.6 Luna	.77	12.8	1.96	45	fail(32)	pass	0
Qwen3.5-397B	.74	13.2	5.43	45	fail(19)	pass	0
Gemini 3.1 Pro	.73	13.6	2.12	44	fail(20)	pass	0
Claude Sonnet 4.6	.77	15.2	1.17	45	fail(6)	pass	0
Claude Opus 4.7	.74	15.5	1.25	36	pass	pass	0
GPT-5.4 Mini	.41	19.4	5.51	45	fail(100)	pass	99
DeepSeek-V4-Pro	.70	19.7	1.81	45	pass	pass	0
Claude Opus 5	.74	14.1	–	–	–	–	0
GPT-5.6 Terra	.75	14.4	–	–	–	–	0

Table 10. First iteration, within-judge repeat SD with the zero-shot sheet and the holistic instrument on the same 45 answers, 3 passes each (stored totals). 95% CIs from an exam-clustered bootstrap; sign test on answers. The pooled estimate averages the capable judges without weights; no correction for multiple comparisons.

Judge	Sheet	Holistic	Δ	95% CI	Sign p	Pooled
GPT-5.6 Luna	1.97	2.25	−0.28	−0.76 to +0.16	.65	yes
Claude Sonnet 4.6	1.19	0.83	+0.36	+0.10 to +0.65	.53	yes
Gemini 3.1 Pro	2.14	2.42	−0.28	−0.83 to +0.28	.88	yes
DeepSeek-V4-Pro	1.81	1.67	+0.13	−0.48 to +0.65	1.00	yes
Qwen3.5-397B	5.43	5.22	+0.21	−3.08 to +3.80	.37	yes
GPT-5.4 Mini	5.52	3.87	+1.65	−0.17 to +3.46	.04	no
Pooled (unweighted)			+0.03	−0.62 to +0.73		

Table 11. First iteration, sheet sources ranked by outcome under Luna: agreement, MAE, signed bias, repeat SD, and the within-exam paired Δ MAE (negative: better than the exam’s average source), with the bootstrap median rank and its 95% interval. Right: MAE and bias under a single-pass Sonnet lens.

Source	Sheets	r	MAE	Bias	SD_{rep}	Δ MAE	Rank	MAE_S	$Bias_S$
Gemini 3.1 Flash-Lite	15	.82	11.4	+2.0	2.40	−2.4	2 [1–10]	13.7	−5.9
Qwen3-235B-Thinking	15	.81	12.0	−5.5	2.19	−1.9	3 [1–8]	13.2	−11.2
Qwen3.6-35B	15	.78	12.5	−4.9	2.21	−1.4	4 [1–8]	13.8	−10.2
GPT-5.4	15	.80	12.6	−6.2	1.80	−1.2	5 [2–8]	14.7	−10.8
Qwen3.5-122B	15	.76	12.8	+1.3	1.73	−1.1	5 [1–10]	14.1	−5.8
Gemini 3.1 Pro	15	.75	14.4	−9.0	1.97	+0.6	8 [2–12]	16.0	−13.1
Claude Opus 4.7	15	.79	14.7	−11.0	1.86	+0.9	8 [4–11]	17.7	−17.0
Claude Sonnet 4.6	15	.81	15.5	−14.5	1.70	+1.7	10 [5–12]	17.8	−17.4
DeepSeek-V4-Pro	15	.79	15.6	−11.6	1.93	+1.8	10 [7–12]	18.7	−16.4
DeepSeek-V4-Flash	15	.40	16.4	−3.5	3.18	+2.6	11 [4–12]	16.0	−12.2
<i>Incomplete coverage (conditional on a valid sheet):</i>									
GPT-5.4 Mini	8	.88	10.3	−0.5	1.66	−3.2	2 [1–8]	12.4	−5.6
Llama-4 Maverick	12	.78	15.9	+9.8	4.07	+2.7	11 [5–12]	11.8	+1.0

Table 12. First iteration, probe battery by arm and judge: negative-battery verdict (fail(N) = failing passes), positive control, and the median/p95/max score per probe type. Criteria on every pass: repetition < 5, empty = 0, off-topic $\leq$ 10, model solution $\geq$ 70.

Arm, judge	Battery	Pos.	Repetition	Empty	Off-topic	Model sol.
Zero-shot, Claude Opus 4.7	pass	pass	0/0/0	0/0/0	0/1/4	99/100/100
Zero-shot, Claude Sonnet 4.6	fail(6)	pass	0/100/100	0/24/81	0/0/0	100/100/100
Zero-shot, DeepSeek-V4-Pro	pass	pass	0/0/0	0/0/0	0/0/0	100/100/100
Zero-shot, GPT-5.4 Mini	fail(100)	pass	100/100/100	95/100/100	8/83/94	100/100/100
Zero-shot, GPT-5.6 Luna	fail(32)	pass	0/100/100	89/100/100	0/1/5	100/100/100
Zero-shot, Gemini 3.1 Pro	fail(20)	pass	80/100/100	100/100/100	0/0/0	100/100/100
Zero-shot, Qwen3.5-397B	fail(19)	pass	100/100/100	79/100/100	0/0/0	100/100/100
Holistic, GPT-5.4 Mini	fail(125)	fail(13)	45/69/71	5/5/7	13/21/22	82/91/95
Holistic, GPT-5.6 Luna	fail(17)	pass	0/2/93	0/0/0	8/18/19	98/100/100
Few-shot, GPT-5.4 Mini	fail(96)	−	99/100/100	95/100/100	5/89/95	−
Few-shot, GPT-5.6 Luna	fail(31)	−	0/100/100	90/100/100	0/0/0	−

Table 13. Apportionment of the 170 first-iteration sheets under three routines: steps outside their fair range, sheets affected, order-inversion pairs, pairs of distinct weights that receive equal points, median relative rounding error, and steps whose points differ from the as-run result.

Routine	Steps	Sheets	Inversions	Collapsed	Rel. error	Changed
As run (first iteration)	27	11	12	458	0.033	0
Hamilton, floor first (integer)	0	0	0	286	0.033	54
Hamilton, floor first (half points)	0	0	0	0	0.020	1060

C Apportionment audit

The audit re-derives the points of all 170 first-iteration sheets (2542 steps) from their stated weights under each routine of Table 13. A step lies outside its fair range when its points differ from its quota by a full unit or more; an order inversion is a pair in which the lighter step receives more points. The as-run routine is reproduced exactly by a copy of the original code.

Table 14. Weight sensitivity of the first-iteration sheets: agreement r with the human pool under five weight schemes, the share of within-sheet permutations whose r falls below the actual one, and repeat SD under the actual and uniform weights.

Judge	Actual	Model	Class	Uniform	$\sqrt{\text{pts}}$	Perm.	SD act.	SD unif.
Claude Opus 4.7	.741	.738	.730	.724	.738	88%	1.25	1.15
Claude Sonnet 4.6	.772	.771	.765	.758	.769	87%	1.17	1.32
DeepSeek-V4-Pro	.700	.697	.683	.691	.699	75%	1.81	1.61
GPT-5.4 Mini	.411	.410	.383	.378	.398	92%	5.51	4.85
GPT-5.6 Luna	.769	.767	.764	.724	.754	99%	1.96	1.77
Gemini 3.1 Pro	.732	.728	.724	.698	.722	96%	2.12	1.90
Qwen3.5-397B	.745	.742	.731	.716	.738	93%	5.43	5.05

Table 15. Steps of 15+ points in the first-iteration sheets: answers and step cells, share of points, share of summed per-step repeat variance, and their ratio.

Judge	Answers	Step cells	Points	Variance	Ratio
DeepSeek-V4-Pro	45	45	22.4%	65.0%	2.90
Claude Opus 4.7	36	21	15.0%	40.7%	2.71
Gemini 3.1 Pro	44	44	22.6%	42.7%	1.89
GPT-5.6 Luna	45	45	22.4%	36.0%	1.61
Claude Sonnet 4.6	45	45	22.4%	32.5%	1.45
GPT-5.4 Mini	45	45	22.4%	28.4%	1.27
Qwen3.5-397B	45	45	22.4%	25.5%	1.14

D Weight sensitivity

Every first-iteration judgment stores a score and a maximum per step. Re-weighting the stored fractions shows how much the generated allocation contributes to agreement and stability without new judge calls (Table 14). *Actual* reproduces the step sums; *class* weights each step by its weight class (routine 1, load-bearing 2, focal 3); *uniform* counts every step the same; *permuted* shuffles a sheet’s own points across its steps. The judges saw the actual maxima, so every scheme is conditional on them.

E Step size and variance

For every judge with at least 3 passes per answer, we sum the per-step variance across passes by step size and compare it with the share of points in that size class (Table 15). The share ignores covariance between steps, so it is a share of *summed* step variance. We call a size class disproportionate when its share of variance exceeds its share of points by more than a factor of 1.3.

F Precision

The minimum detectable effect of a paired contrast is $(t_{1-\alpha/2, n-1} + t_{\text{power}, n-1}) \sigma_d / \sqrt{n}$. We take σ_d from the closest existing contrast, Luna with the zero-shot sheet vs the holistic instrument on D1, so the values are proxies until the new instrument’s data exist (Table 16). The D2 proxy uses a holistic human reference, which is noisier than per-step expert grades.

G Probe battery

The probe battery runs per judge and instrument over 3 passes before any benchmark run (Table 17). A judge that fails a criterion on any pass is not used with that instrument. An alternative-path

Table 16. Minimum detectable effects at power .8 (t -based, paired). σ_d : SD of the paired differences in the proxy contrast.

Contrast	n	σ_d	$\alpha = .025$	$\alpha = .05$
Repeat SD, D1	45	1.87	0.88	0.80
Repeat SD, D1, 5 passes (optimistic)	45		0.63	0.57
Absolute error, D2a (15 scripts)		12.4	10.9	9.7
Absolute error, all distinct D2 scripts (38)		12.4	6.4	5.8
Absolute error, D2, dependent pair once (37)		12.4	6.5	5.9

Table 17. Probe battery of the extended study. Totals are out of 100 BE.

Probe	Input	Criterion
empty	a near-empty answer	total = 0
repetition	the case text, verbatim	total < 5
off-topic	the model solution of another exam	total $\leq$ 10
same-area off-topic	the model solution of another exam in the same area	total $\leq$ 10
keyword salad	norms and key terms of the sheet, no sentences	total $\leq$ 10
injection	an answer that instructs the judge to award full marks	total = 0
model solution	the exam’s own model solution	total $\geq$ 70 (descriptive for the rating unit)
section ablation	the model solution with one Abschnitt removed	drop close to the removed Abschnitt’s BE
misplacement	correct reasoning moved under another heading	drop close to 0
negation flip	result sentences of the model solution negated	descriptive

probe follows in E5: the model solution rewritten along one alternative path and checked by the exam author.

H Verifier rules

The verifier (`services/workers/ml_evaluation/llm_judge_evaluator.py`) accepts a quote only if every fragment passes:

- *Normalization*. Unicode normalization; literal line-break escapes, markup escapes, footnote marks, emphasis markers and inline HTML removed; quotation marks and dashes unified; the result marks “(+)” and “(-)” turned into the words *positiv* and *negativ*; case folded; white-space collapsed.
- *Fragments*. A quote is split at ellipses. Each fragment needs at least 3 tokens, or two tokens with 15 letters together, and more than a norm citation. The whole quote needs 4 word characters.
- *Matching*. Fragments with fewer than 4 tokens must match on word boundaries. Longer ones may match as a substring or token by token in order, with up to 2 skipped answer words between quoted words and one missed quoted word per 8. Unprotected words of at least 5 letters may differ by one edit or a small inflection. Glued passages without an ellipsis are checked sentence by sentence.
- *Protected tokens*. Negations, result words and their antonyms (including every *un-* form), qualifiers such as *nur* and *teilweise*, Roman numerals and every token with a digit must match exactly. They may not be dropped, added, swapped or stepped over. Numbers never match part of a longer number.

Table 18. Planned cost in USD per judge call (about 25k input and 6k output tokens) and per generated sheet, from pilot-measured tokens at catalog prices. Open-model sheets cost less than the value shown. [PENDING: measured costs]

Model	USD
GPT-5.6 Luna	0.013
GPT-5.4 Mini	0.046
DeepSeek-V4-Pro	0.070
Gemini 3.1 Pro	0.120
GPT-5.4	0.150
Claude Sonnet 4.6	0.170
Sheet: open models	0.05
Sheet: GPT-5.4 Mini	0.15
Sheet: GPT-5.4	0.60
Sheet: Claude Opus 4.7	1.00

- *Edges*. A match fails if a negation or a condition (*ob*, *wenn*, *falls*, *sofern*, *soweit*) stands right before it in the same clause, or a negation, a negative result mark or a condition right after it in the same sentence.

I Validator checks

Strict checks fail an attempt and return their messages to the model: the document structure and every cross-reference; steps above 10 BE; steps above 3 BE with a single bullet; bullets that can be met by an absence or that are conditional; steps named as a result; forks whose steps depend on replaced steps; a fork that replaces a whole section; and a path whose bullets repeat those of the replaced steps (token overlap of at least 0.6). *Soft* checks only log hints: broader absence patterns; forks that replace exactly one step with one step; forks above 20 BE; near-duplicate bullets; bare naming bullets shorter than 8 words; a sheet on which most steps exceed 4 BE; and rescaled sections in top-down allocation.

J Designs for alternative solution paths

A dispute that changes only a result or its reasons (a local alternative) stays inside a step as an outcome-neutral bullet, such as “discusses the dispute with the arguments of both sides; both results tenable”. A structural fork replaces named steps of the model solution’s path with the steps of each alternative path, within the same budget. We compare three designs (E5):

- *A0, replace*. Only the model solution’s path is scored. The rendered sheet tells the judge to credit work on another tenable path on the replaced steps. This is the first iteration’s design; it needs the step or rating unit, because a bullet of a replaced step describes that step’s own requirement.
- *A1, branch, best path*. The judge scores the steps of every path; code counts the path with the most points.
- *A2, branch, declared path*. As A1, but code counts the path the judge declares the answer follows, backed by a verified quote; without one it counts the model solution’s path.

A1 and A2 come from the same call. A path counts as followed only if the answer argues it as its tenability requirements demand; a bare other result does not count.

K Costs

The full program is planned at USD 430 to USD 480 (Table 3); a lean version with fewer generators and judges at about USD 200.

L Data statement

D1. The first-iteration corpus of 15 German exams and 45 answers with blind human grades [34]. Exam texts are third-party material used under the permissions documented for that corpus and are not redistributed. Human grades are pseudonymized.

D2. One police-law exam with its author's sheet and key, 15 graded scripts (D2a) and 23 student submissions (D2b), used with the students' consent for research. All texts are pseudonymized before any research call to a model provider. The material stays private; only aggregate statistics are published.

Z2. Gregor Roth's grading sheet and workbook for a civil-law exam, without case, solution or scripts. Private; described only by its structure.

Model calls. Every call records the model id, parameters, prompt hashes and the code version. For D1 we release aggregates; the per-call records are withheld for size. [PENDING: owner: release the per-call records with the dataset release?]

M Output specification of the checklist sheet

[PENDING: the verbatim output specification (German) and its English translation, after the E0 freeze]

N Example generated sheet

[PENDING: one generated checklist sheet for a D1 exam, as rendered for the judge, after the E0 freeze; D1 only, never D2]